Participant Information and Informed Consent Form

Study title	MR-Driven Virtual Fixture Generation and Physical Human-Robot Interaction Study
Principal investigator	
Institution	
Study site	
Researcher contact	
Ethics approval	
Estimated Duration	Approximately 120 minutes

Please read this form carefully before deciding whether to participate. You may ask questions at any time. Taking part is voluntary. If you decide to participate, you will be asked to sign the consent section at the end of this document.

1. Why is this study being conducted?

This study investigates ways to make it easier for people to create, adjust, and use motion guidance for collaborative robots. The research combines mixed reality, motion demonstration, and virtual fixtures. A virtual fixture is a form of software-generated guidance that can help a person move a robot along a desired path.

The study compares several methods for specifying robot paths and several forms of guidance during a physical human-robot interaction task. We are interested in task performance, workload, and usability.

2. Why am I being invited?

You are being invited as a volunteer participant in a user study of human-robot interaction. Before the study begins, the researcher will confirm that you meet the approved eligibility requirements.

3. What will happen if I take part?

If you agree to participate, the study will include the following activities:

- You will receive an explanation of the study equipment, safety procedures, and experimental task.
- You will receive hands-on familiarization and practice with the required interactions.
- You will complete a steady-hand task using a collaborative robot. A ring-shaped robot end-effector must be guided along a spatial wire while avoiding contact.
- You will experience several path-specification methods. These include a screen-based interface, a tracked handheld demonstration method, and mixed-reality path generation using a Meta Quest 3 headset. In one mixed-reality condition, you will also be able to edit the generated path.
- You will perform the physical task under different guidance conditions, including no virtual-fixture guidance, guidance generated from different paths, and a condition that provides translational guidance without rotational guidance.
- The order of experimental conditions will be randomized.
- For selected conditions, you will complete the NASA Task Load Index (NASA-TLX) and System Usability Scale (SUS). You may also provide brief written comments about workload, usability, difficulties, and suggested improvements.
- You may request a break whenever needed.

Estimated total time: approximately 120 minutes.

4. What information will be collected?

The study may record information such as the paths you create, robot motion, task completion time, path-following errors, contact with the task wire, and your questionnaire responses. Written comments may also be collected if you choose to provide them.

5. What are the possible risks, and what happens if I feel unwell or am injured?

This study involves a collaborative robot and, in some conditions, a head-mounted mixed-reality display. Possible risks and discomforts include:

- Temporary eye strain, headache, dizziness, nausea, visual fatigue, or disorientation while using the mixed-reality headset.
- Temporary arm, hand, shoulder, or general fatigue from repeated interaction with the robot or interfaces.
- Minor accidental contact with the robot or task apparatus. Unexpected motion is possible if equipment or tracking does not behave as intended.
- Temporary frustration, mental workload, or performance pressure while completing precise tasks.

The research team will check the equipment before the session, keep the experimental area controlled, monitor you during robot interaction, and keep an emergency stop available.

If you feel unwell or uncomfortable at any time, tell the researcher immediately. You may remove the mixed-reality headset, request a break, or stop the study. The researcher may also pause or stop a task if there is a safety concern.

Medical treatment, compensation for research-related injury, and relevant institutional procedures: *Anonymized*

6. Are there any benefits?

You may not receive a direct personal benefit from participating. Your participation may help researchers understand how mixed reality and haptic guidance can be designed to make collaborative robot interaction more intuitive and effective.

7. Will I be paid, and will participation cost me anything?

No payment or reimbursement will be provided for participation in this study.

There is no charge for taking part in this study.

8. How will my information be protected?

Your research data will be labeled with a participant ID whenever possible. Direct identifiers will be kept separate from the research data and will not be included in publications or presentations.

Research results may be published or presented in academic venues. Only processed, de-identified, or aggregated results will be reported or shared. Raw identifiable participant data will not be published.

9. Is participation voluntary, and what happens if I withdraw?

Taking part is entirely voluntary. You may choose not to participate. If you participate, you may skip any optional question, request a break, or stop the study at any time without giving a reason and without penalty.

If you withdraw from the study, the study procedures will stop. Any data collected from you during this study will be excluded from the research analysis and will not be reported in the study results.

If you stop because of discomfort or an equipment-related issue, the researcher may ask whether you need assistance and will document any safety incident according to institutional procedures.

Consent Statement

By signing below, I confirm that:

☐ I have read and understood the information above.

☐ I have had the opportunity to ask questions and have received satisfactory answers.

☐ I understand the study procedures, including interaction with a collaborative robot and the use of a mixed-reality headset in some conditions.

☐ I understand the possible risks and discomforts associated with participation.

☐ I understand that participation is voluntary and that I may stop at any time without penalty.

☐ I understand that, if I withdraw, my study data will be excluded from the research analysis and will not be reported in the study results.

☐ I understand that my study data will be handled confidentially and that only processed, de-identified, or aggregated results may be reported or shared.

☐ I voluntarily agree to participate in this study.

Participant name	
Participant signature	
Date	
Researcher name	
Researcher signature and date	